

Session 2 — Linear algebra II

Solutions

As in Session 1, these solutions are written to be read *on their own*. Each exercise opens with a shaded box recalling the one definition or formula it rests on, and why that formula is what it is; then the computation is carried out with every intermediate step visible; then the result is read in words. Many exercises end by naming the mistake the question was built to catch. Everything here is one question: *how many directions does this table really have?* — answered first algebraically (rank), then dynamically (eigenvalues), then statistically (PCA), then in the only form that works for a non-square table (the SVD).

Four formulas used constantly below. For a 2×2 matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\det \mathbf{A} = ad - bc, \quad \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

$$T = \operatorname{tr} \mathbf{A} = a + d = \lambda_1 + \lambda_2, \quad D = \det \mathbf{A} = \lambda_1 \lambda_2, \quad \lambda^2 - T\lambda + D = 0.$$

The last line is the whole of the eigenvalue computation in 2×2 : write down two numbers you can read off the matrix, solve one quadratic.

Exercise 1. Compute (a) $\det\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$ and its inverse; (b) $\det\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$ and its inverse; (c) $\det\begin{pmatrix} 4 & 6 \\ 2 & 3 \end{pmatrix}$ — invertible? rank?

What this exercise uses. The determinant of a 2×2 matrix is $ad - bc$; geometrically (Session 1, Exercise 25) its absolute value is the factor by which areas are multiplied, and its sign says whether the plane was flipped. The inverse exists **exactly when** $\det \neq 0$, and then $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$: swap the diagonal entries, flip the sign of the other two, divide by the determinant. The reason $\det = 0$ blocks inversion is geometric, not algebraic: area factor 0 means the plane is squashed onto a line, and once two distinct inputs share an output there is no way back.

Solution.

(a) $\det = 5 \cdot 4 - 2 \cdot 3 = 20 - 6 = 14 \neq 0$, so the inverse exists:

$$\mathbf{A}^{-1} = \frac{1}{14} \begin{pmatrix} 4 & -2 \\ -3 & 5 \end{pmatrix} = \begin{pmatrix} \frac{2}{7} & -\frac{1}{7} \\ -\frac{3}{14} & \frac{5}{14} \end{pmatrix}.$$

Check on the first column — always do it:

$$\frac{1}{14} \begin{pmatrix} 4 & -2 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 20 - 6 \\ -15 + 15 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 14 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \checkmark$$

(b) $\det = 2 \cdot 3 - 5 \cdot 1 = 6 - 5 = 1$. The determinant is 1, so the division does nothing and the inverse is read off directly:

$$\mathbf{A}^{-1} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}.$$

Check: $\begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 6-5 & 15-15 \\ -2+2 & -5+6 \end{pmatrix} = \mathbf{I}. \checkmark$

(c) $\det = 4 \cdot 3 - 6 \cdot 2 = 12 - 12 = 0$. **Not invertible.** The rank is 1: it is not 0 (the matrix is not the zero matrix, so it has at least one independent column), and it is not 2 (the determinant vanishes, so the two columns are dependent — indeed $\begin{pmatrix} 6 \\ 2 \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 4 \\ 2 \end{pmatrix}$). Every output lies on the single line spanned by $(4, 2)$, i.e. by $(2, 1)$.

Common mistake. Writing $\frac{1}{ad-bc} \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}$ — forgetting that a and d are *exchanged*, not left in place. The check on one column costs five seconds and catches it every time.

Exercise 2. (a) For which k is $\begin{pmatrix} 2 & 3 \\ 4 & k \end{pmatrix}$ non-invertible, what happens to the rank there, and how is the critical matrix an outer product? (b) Same question for $\begin{pmatrix} 1 & 4 \\ 3 & k \end{pmatrix}$.

What this exercise uses. “Non-invertible” and “ $\det = 0$ ” are the same condition, so a question of the form “for which $k \dots$ ” is a single equation in k . And when the determinant of a 2×2 matrix vanishes, the two columns are proportional, which lets the whole matrix be written as an **outer product** $\mathbf{mn}^T$: one column direction $\mathbf{m}$, one list of multipliers $\mathbf{n}$. Read as a table: everyone has the same profile, only the level differs.

Solution.

(a) $\det = 2k - 3 \cdot 4 = 2k - 12$, which vanishes for $\boxed{k = 6}$. For every $k \neq 6$ the rank is 2 (full); at $k = 6$ it **drops to** 1. Note that it drops to 1 and not to 0: a matrix loses full rank at $\det = 0$, it does not collapse entirely.

At $k = 6$ the matrix is $\begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix}$, whose second column is $\frac{3}{2}$ times the first. Take the first column as the direction and read the multipliers off the rows:

$$\begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \begin{pmatrix} 1 & \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 2 & 3 \end{pmatrix}.$$

Check the second form: $1 \cdot (2, 3) = (2, 3)$ is row 1 and $2 \cdot (2, 3) = (4, 6)$ is row 2. ✓ The decomposition is not unique — $\mathbf{m}$ and $\mathbf{n}$ can be rescaled inversely — but the two *directions* are.

(b) $\det = 1 \cdot k - 4 \cdot 3 = k - 12$, so $\boxed{k = 12}$, and again the rank drops from 2 to 1 there ($\begin{pmatrix} 1 & 4 \\ 3 & 12 \end{pmatrix}$ has second column = $4 \times$ first).

Remark. This is the algebraic version of a very concrete data situation: a table has full rank for almost every value of a parameter, and loses rank on a thin exceptional set. Exercise 24 is about what happens when a real, noisy table sits near that set without being on it.

Exercise 3. (a) $\det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1$: what do the size and the sign mean geometrically? (b) Check $\det(AB) = \det A \det B$ for $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

What this exercise uses. $|\det|$ is the area factor and the sign is the orientation. The multiplicativity $\det(AB) = \det A \det B$ is then obvious rather than mysterious: AB means “do B , then A ”, and if B multiplies areas by $|\det B|$ and A by $|\det A|$, then doing both multiplies them by the product. The signs multiply for the same reason — two flips undo each other.

Solution.

(a) **Size** $|-1| = 1$: areas are preserved. Nothing is stretched or shrunk.

Sign $-$: the orientation of the plane is reversed. The matrix leaves $(1, 0)$ alone and sends $(0, 1)$ to $(0, -1)$: it is the **reflection in the x -axis**. A mirror preserves every area and every length while exchanging left and right — and no rotation can do that, since rotations have determinant $+1$.

(b) Separately:

$$\det A = 2 \cdot 3 - 0 = 6, \quad \det B = 0 \cdot 0 - 1 \cdot 1 = -1.$$

The product:

$$AB = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 \cdot 0 + 0 \cdot 1 & 2 \cdot 1 + 0 \cdot 0 \\ 0 \cdot 0 + 3 \cdot 1 & 0 \cdot 1 + 3 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 3 & 0 \end{pmatrix}, \quad \det(AB) = 0 - 6 = -6.$$

And $\det A \cdot \det B = 6 \times (-1) = -6$. ✓ In words: swapping the axes flips the plane over $(-)$, stretching by 2 and 3 multiplies areas by 6; do both and areas are multiplied by 6 with the orientation reversed.

Remark. One consequence used repeatedly later: if $\det A \neq 0$ and $\det B \neq 0$ then $\det(AB) \neq 0$ — invertible composed with invertible is invertible. And $\det(A^{-1}) = 1/\det A$, since $\det(A)\det(A^{-1}) = \det(\mathbf{I}) = 1$.

Exercise 4. (a) Rank of $\begin{pmatrix} 2 & 6 \\ 3 & 9 \end{pmatrix}$; inverse? on what line do the outputs lie? (b) Rank of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$. (c) A 2×2 matrix of rank 0, one of rank 1, one of rank 2.

What this exercise uses. The **rank** of a matrix is the number of *independent* columns — equivalently (a theorem, not an accident) the number of independent rows, and equivalently the dimension of the set of all outputs $\{\mathbf{Ax}\}$, which is $\text{span}(\text{columns})$. Three consequences used below: $\text{rank} \leq \min(\text{rows}, \text{columns})$; a square matrix is invertible iff its rank is full; and $\text{rank} = 0$ happens only for the zero matrix, since a single nonzero column already gives rank 1.

Solution.

- (a) $\det = 2 \cdot 9 - 6 \cdot 3 = 18 - 18 = 0$, so the rank is not 2; the matrix is nonzero, so the rank is not 0. **Rank 1**, hence **no inverse**. The outputs: since $\mathbf{Ax} = x_1 \begin{pmatrix} 2 \\ 3 \end{pmatrix} + x_2 \begin{pmatrix} 6 \\ 9 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ 9 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 3 \end{pmatrix}$,

$$\mathbf{Ax} = (x_1 + 3x_2) \begin{pmatrix} 2 \\ 3 \end{pmatrix},$$

so *every* output lies on the line $\text{span}\{(2, 3)\}$ — and every point of that line is reached (take $x_2 = 0$). The plane has been squashed onto a line; that is what rank 1 means in a picture.

- (b) The second row is exactly twice the first, so the two rows are dependent and the rank is not 2; the matrix is nonzero, so **rank 1**. The columns confirm it: $(2, 4) = 2(1, 2)$ and $(3, 6) = 3(1, 2)$, so all three columns lie on one line. Consistency check with the bound: $\text{rank} \leq \min(2, 3) = 2$, and we found 1. ✓

- (c) For instance

$$\underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}}_{\text{rank 0}}, \quad \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{\text{rank 1}}, \quad \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{\text{rank 2}}.$$

Rank 0 forces *all* entries to vanish: one nonzero entry already gives a nonzero column, hence rank at least 1. Any matrix with $\det \neq 0$ works for rank 2; any nonzero matrix with $\det = 0$ works for rank 1.

Common mistake. Reading “ $\det = 0$ ” as “the matrix is zero”, or as “rank 0”. It says the rank is *not full* — for 2×2 that leaves both 1 and 0, and 0 occurs only in the single case of the zero matrix.

Exercise 5. X records item 1, item 2 and the difference item 1 — item 2: $X = \begin{pmatrix} 4 & 9 & -5 \\ 6 & 7 & -1 \\ 5 & 8 & -3 \\ 9 & 4 & 5 \\ 7 & 6 & 1 \end{pmatrix}$. (a) A vanishing combination of the columns, and the rank. (b) $X\mathbf{w} = X\mathbf{w}'$ for $\mathbf{w} = (1, -1, 0)$, $\mathbf{w}' = (0, 0, 1)$; a third such weight vector. (c) What does that mean for a regression on all three columns?

What this exercise uses. $X\mathbf{w} = w_1\mathbf{c}_1 + w_2\mathbf{c}_2 + w_3\mathbf{c}_3$: a matrix–vector product is a combination of the *columns* with the entries of $\mathbf{w}$ as coefficients (Session 1, Exercise 25). Hence a vanishing combination of columns is precisely a nonzero $\mathbf{z}$ with $X\mathbf{z} = \mathbf{0}$; the set of such $\mathbf{z}$ is the **kernel**. Two consequences, and they are the same statement twice: $\text{rank } X < (\text{number of columns})$, and the map $\mathbf{w} \mapsto X\mathbf{w}$ is *not injective* — $X(\mathbf{w} + \mathbf{z}) = X\mathbf{w} + X\mathbf{z} = X\mathbf{w}$ for every $\mathbf{z}$ in the kernel.

Solution.

- (a) By construction column 3 is column 1 minus column 2. Check it on every row:

$$4 - 9 = -5, \quad 6 - 7 = -1, \quad 5 - 8 = -3, \quad 9 - 4 = 5, \quad 7 - 6 = 1. \checkmark$$

So $\mathbf{c}_1 - \mathbf{c}_2 - \mathbf{c}_3 = \mathbf{0}$, with coefficients $(1, -1, -1)$, not all zero: the three columns are **dependent** and $\text{rank } X \leq 2$.

Why the rank is not smaller. Rank ≤ 1 would mean $\mathbf{c}_2$ is a multiple of $\mathbf{c}_1$. The first row would force $\lambda = 9/4$, the second $\lambda = 7/6$; these differ, so no such λ exists. Two independent columns, therefore

$$\boxed{\text{rank } X = 2}.$$

Read in words: the table has *three* columns but only *two* directions. The third measure is a bookkeeping consequence of the first two and carries no new information whatsoever.

- (b)

$$X\mathbf{w} = \mathbf{c}_1 - \mathbf{c}_2 = (4 - 9, 6 - 7, 5 - 8, 9 - 4, 7 - 6) = (-5, -1, -3, 5, 1),$$

$$X\mathbf{w}' = \mathbf{c}_3 = (-5, -1, -3, 5, 1).$$

The same vector. $\checkmark$ This is not a coincidence to be verified but a restatement of (a): $\mathbf{w} - \mathbf{w}' = (1, -1, -1)$ is exactly the vanishing combination, so $X(\mathbf{w} - \mathbf{w}') = \mathbf{0}$.

A third: add any multiple of $(1, -1, -1)$. With $t = 1$,

$$\mathbf{w}'' = (1, -1, 0) + (1, -1, -1) = (2, -2, -1), \quad X\mathbf{w}'' = 2\mathbf{c}_1 - 2\mathbf{c}_2 - \mathbf{c}_3 = 2\mathbf{0} + (\mathbf{c}_1 - \mathbf{c}_2 - \mathbf{c}_3) + \dots$$

— more directly, $X\mathbf{w}'' = X\mathbf{w} + X(1, -1, -1) = X\mathbf{w} + \mathbf{0} = (-5, -1, -3, 5, 1)$. $\checkmark$ In fact there is a whole *line* of weight vectors giving this same prediction: $\mathbf{w} + t(1, -1, -1)$ for every real t .

- (c) The coefficients are **not identifiable**. Infinitely many weight vectors produce identical predictions for every participant, so the data cannot single one out: there is no such thing as “the effect of item 1” here. This is exact **collinearity**, and no sample size fixes it — the remedy is to drop the redundant column.

Common mistake. Expecting the software to fail. Some routines return an answer without complaint (or a warning easily missed), and the returned coefficients are then one arbitrary point on a line of equally good ones. The fit and the predictions are fine; it is the *interpretation of individual coefficients* that is meaningless.

Exercise 6. Compute the outer product $\begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 \end{pmatrix}$, then its determinant and its rank.

What this exercise uses. An **outer product** $\mathbf{m}\mathbf{n}^\top$ multiplies a column $(k \times 1)$ by a row $(1 \times p)$ — inner dimensions 1 and 1 agree — and returns a $k \times p$ matrix whose entry (i, j) is $m_i n_j$. Do not confuse it with the dot product $\mathbf{m}^\top \mathbf{n}$, which is $(1 \times k)(k \times 1) = 1 \times 1$: one number. Same two vectors, opposite order, completely different objects.

Solution. Entry (i, j) is $m_i n_j$:

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 3 & 1 \cdot (-1) \\ 2 \cdot 3 & 2 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 6 & -2 \end{pmatrix}.$$

Determinant:

$$\det = 3 \cdot (-2) - (-1) \cdot 6 = -6 + 6 = 0.$$

Rank 1: the matrix is nonzero, and both columns are multiples of $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ — column 1 is $3\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, column 2 is $-1\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

Why this had to happen, with no computation. Every column of $\mathbf{m}\mathbf{n}^\top$ is $n_j\mathbf{m}$: one single direction $\mathbf{m}$, rescaled. So a (nonzero) outer product *always* has rank exactly 1 — and conversely every rank-1 matrix can be written this way. *Rank 1 and “outer product” are the same thing*, which is why $\mathbf{m}\mathbf{n}^\top$ will be the building block of the SVD in Exercises 22–24.

Common mistake. Computing $\begin{pmatrix} 3 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 3 - 2 = 1$ instead. That is the *inner* product; the order of the two factors decides which of the two objects you get, and only one of them is a matrix.

Exercise 7. (a) A 200×20 table is exactly $X = \mathbf{m}\mathbf{n}^\top$: rank? how many numbers to store instead of 4000? (b) $X = \mathbf{m}_1\mathbf{n}_1^\top + \mathbf{m}_2\mathbf{n}_2^\top$ with $\mathbf{m}_1, \mathbf{m}_2$ independent and $\mathbf{n}_1, \mathbf{n}_2$ independent: rank, and what does the table look like?

What this exercise uses. Ranks add at most: $\text{rank}(A + B) \leq \text{rank } A + \text{rank } B$, because every column of $A + B$ is a sum of a column of A and a column of B , so all of them live in the sum of the two column spans. When the two pieces are “genuinely different” — here, the $\mathbf{m}$ ’s independent and the $\mathbf{n}$ ’s independent — the bound is attained. And a rank- r table of size $n \times p$ is stored as r pairs (column of length n , row of length p), i.e. $r(n + p)$ numbers instead of np .

Solution.

(a) **Rank 1** (Exercise 6). Storage: the vector $\mathbf{m}$ (200 numbers) and the vector $\mathbf{n}$ (20 numbers), so

$$200 + 20 = 220 \quad \text{numbers instead of} \quad 200 \times 20 = 4000,$$

a factor of about 18. Nothing is lost: from $\mathbf{m}$ and $\mathbf{n}$ every one of the 4000 entries is recomputed as $m_i n_j$.

Reading: all 200 participants share *one and the same* response profile $\mathbf{n}$ across the 20 items; the only thing distinguishing person i is a single overall level m_i . One number per person is the entire inter-individual variability.

(b) **Rank 2.** Each row of X is

$$\text{row } i = m_{1i} \mathbf{n}_1^\top + m_{2i} \mathbf{n}_2^\top,$$

a mixture of exactly *two* fixed profiles $\mathbf{n}_1, \mathbf{n}_2$, with person i ’s own two mixing coefficients (m_{1i}, m_{2i}) . The rank is not 1 because $\mathbf{n}_1, \mathbf{n}_2$ are independent (two genuinely different profiles are needed), and not 3 or more because only two were used.

Storage: $2(200 + 20) = 440$ numbers. In the language of factor analysis: two latent components, two loadings per person.

Remark. Exercises 22–24 are this exercise run in reverse. There the table is given and the decomposition $X = \sum_k \sigma_k \mathbf{u}_k \mathbf{v}_k^\top$ must be found; the SVD is the machine that finds it, and it does so in the best possible order — largest piece first.

Exercise 8. (a) Verify that $(2, 1)$ is an eigenvector of $\begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$, give its eigenvalue; is $(4, 2)$ one too? (b) Eigenvalues and eigenvectors of $\begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}$ and of $\begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}$.

What this exercise uses. $\mathbf{v} \neq \mathbf{0}$ is an **eigenvector** of $\mathbf{A}$ with **eigenvalue** λ when $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$: the matrix does not turn this vector, it only rescales it. *Verifying* a proposed eigenvector requires no theory at all — one matrix–vector product and a look. Two structural facts follow immediately from the definition: (i) if $\mathbf{v}$ is an eigenvector so is $c\mathbf{v}$ for every $c \neq 0$, since $\mathbf{A}(c\mathbf{v}) = c\mathbf{A}\mathbf{v} = c\lambda\mathbf{v} = \lambda(c\mathbf{v})$ — an eigenvector is really a *direction*, a whole line; (ii) $\mathbf{v} = \mathbf{0}$ is excluded, since $\mathbf{A}\mathbf{0} = \lambda\mathbf{0}$ holds for every λ and would make the notion empty.

Solution.

(a) One product:

$$\begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \cdot 2 + 2 \cdot 1 \\ 1 \cdot 2 + 3 \cdot 1 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} = 5 \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

The output is a multiple of the input, so $(2, 1)$ is an eigenvector, with $\boxed{\lambda = 5}$.

And $(4, 2) = 2(2, 1)$: same line, so **yes**, it is an eigenvector too, with the *same* eigenvalue 5. Check directly if you like: $\begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 20 \\ 10 \end{pmatrix} = 5 \begin{pmatrix} 4 \\ 2 \end{pmatrix}$. ✓ This is fact (i) above: eigenvectors come in lines, never alone, and that is why software returns them normalised to length 1 (an arbitrary but standard choice of representative).

(b) A **diagonal** matrix scales each axis separately, so the axes are its eigenvectors and the diagonal entries are the eigenvalues:

$$\begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

so $\lambda = 4$ along $(1, 0)$ and $\lambda = -1$ along $(0, 1)$. Likewise $\lambda = 5$ along $(1, 0)$ and $\lambda = -2$ along $(0, 1)$ for the second matrix.

The negative eigenvalue means that direction is not merely shrunk but **reversed**: $(0, 1) \mapsto (0, -1)$.

Remark. This is the whole point of eigenvectors: in a basis made of them, the matrix is diagonal, and a diagonal matrix is trivial to understand and to iterate (Exercise 13). Finding eigenvectors means looking for the axes in which the transformation was simple all along.

Exercise 9. Using T , D and $\lambda^2 - T\lambda + D = 0$: (a) $\begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}$ — T , D , eigenvalues, eigenvectors; (b) $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ — T , D , eigenvalues.

What this exercise uses. λ is an eigenvalue iff $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ has a *nonzero* solution, i.e. iff $\mathbf{A} - \lambda\mathbf{I}$ is not invertible, i.e. iff $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$. For 2×2 , expanding that determinant gives

$$\det \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} = (a - \lambda)(d - \lambda) - bc = \lambda^2 - \underbrace{(a + d)}_T \lambda + \underbrace{(ad - bc)}_D.$$

So the whole characteristic polynomial is determined by two numbers read straight off the matrix. Once λ is known, the eigenvector is found by solving $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ — and since that matrix is singular, its two rows say the same thing: use one, ignore the other.

Solution.

(a) $T = 0 + 0 = 0$ and $D = 0 \cdot 0 - 3 \cdot 3 = -9$, so

$$\lambda^2 - 0 \cdot \lambda + (-9) = 0 \iff \lambda^2 = 9 \iff \lambda = 3 \text{ or } \lambda = -3.$$

For $\lambda = 3$: $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -3 & 3 \\ 3 & -3 \end{pmatrix}$, and the first row gives $-3v_1 + 3v_2 = 0$, i.e. $v_1 = v_2$: the eigenvector is $\boxed{(1, 1)}$ (the second row, $3v_1 - 3v_2 = 0$, says exactly the same — as it must).

For $\lambda = -3$: $\mathbf{A} + 3\mathbf{I} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$ gives $v_1 = -v_2$: the eigenvector is $\boxed{(1, -1)}$.

Checks: $\begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \checkmark$; $\begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \end{pmatrix} = -3 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \checkmark$.

Reading: the matrix swaps the two coordinates and multiplies by 3. It leaves the “sum” direction $(1, 1)$ alone (up to the factor 3) and reverses the “difference” direction $(1, -1)$ — which is what swapping does to a difference. Note also $T = 0 = 3 + (-3)$ and $D = -9 = 3 \times (-3)$. $\checkmark$

(b) $T = 1 + 1 = 2$ and $D = 1 \cdot 1 - (-1) \cdot 1 = 1 + 1 = 2$, so

$$\lambda^2 - 2\lambda + 2 = 0, \quad \Delta = 4 - 8 = -4 < 0, \quad \lambda = \frac{2 \pm \sqrt{-4}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i.$$

Complex, and conjugate to each other — as they must be, the polynomial having real coefficients. Sanity check: $\lambda_1 + \lambda_2 = 2 = T \checkmark$ and $\lambda_1 \lambda_2 = (1 + i)(1 - i) = 1 - i^2 = 2 = D \checkmark$. Geometrically the matrix rotates (see Exercise 11): no real direction is preserved.

Common mistake. Solving $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ by Gaussian elimination and concluding $\mathbf{v} = \mathbf{0}$ “because the system reduces to $0 = 0$ ”. That degenerate line is the *point*: the system is singular by construction, so it has a line of solutions, and any nonzero one is the eigenvector. If you ever get only $\mathbf{v} = \mathbf{0}$, your λ is wrong.

Exercise 10. Eigenvalues and eigenvectors of (a) $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$; (b) $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$; (c) $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$; (d) $\begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$.

What this exercise uses. Same routine four times — T , D , the quadratic, then one row of $\mathbf{A} - \lambda\mathbf{I}$ per eigenvalue. Two shortcuts worth having: for a **triangular** matrix the eigenvalues are the diagonal entries (the characteristic determinant is already triangular, so it is the product of the diagonal terms); and every solution should be checked against $\lambda_1 + \lambda_2 = T$, $\lambda_1 \lambda_2 = D$, which costs one addition and one multiplication.

Solution.

(a) Triangular: the eigenvalues are $\boxed{2}$ and $\boxed{3}$ (check: $T = 5$, $D = 6 \checkmark$).

$\lambda = 2$: $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$, so $v_2 = 0$ and v_1 is free: $\mathbf{v} = (1, 0)$.

$\lambda = 3$: $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$, so $-v_1 + v_2 = 0$: $\mathbf{v} = (1, 1)$.

(b) $T = 2$, $D = 1 - 4 = -3$: $\lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1) = 0$, so $\boxed{3}$ and $\boxed{-1}$.

$\lambda = 3$: $\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$ gives $v_1 = v_2$: $\mathbf{v} = (1, 1)$.

$\lambda = -1$: $\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$ gives $v_1 = -v_2$: $\mathbf{v} = (1, -1)$.

Check: $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} \checkmark$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix} \mapsto \begin{pmatrix} -1 \\ 1 \end{pmatrix} \checkmark$. (This matrix comes back in Exercises 13(c) and 17(c).)

(c) $T = 7$, $D = 12 - 2 = 10$: $\lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0$, so $\boxed{5}$ and $\boxed{2}$.

$\lambda = 5$: $\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}$ gives $v_1 = v_2$: $\mathbf{v} = (1, 1)$.

$\lambda = 2$: $\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$ gives $2v_1 + v_2 = 0$: $\mathbf{v} = (1, -2)$.

Check: $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 4-2 \\ 2-6 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} \checkmark$. Note the eigenvectors here are *not* orthogonal: $(1, 1) \cdot (1, -2) = -1 \neq 0$. Nothing is wrong — the matrix is not symmetric (compare Exercise 15).

(d) Triangular: $\boxed{3}$ and $\boxed{2}$ (check $T = 5$, $D = 6 \checkmark$).

$\lambda = 3$: $\begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}$ gives $v_1 = v_2$: $\mathbf{v} = (1, 1)$.

$\lambda = 2$: $\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ gives $v_1 = 0$: $\mathbf{v} = (0, 1)$.

Remark. (a) and (d) have the same eigenvalues $\{2, 3\}$ but different eigenvectors: eigenvalues alone do not determine a matrix. What determines it is the pair (eigenvalues, eigenvectors) — “what is stretched, and by how much”.

Exercise 11. (a) Eigenvalues of $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ and $\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$: real or complex, and what do they mean geometrically? (b) For which real a does $\begin{pmatrix} 1 & a \\ -a & 1 \end{pmatrix}$ have complex eigenvalues?

What this exercise uses. $\lambda^2 - T\lambda + D = 0$ has discriminant $\Delta = T^2 - 4D$. Real distinct eigenvalues iff $\Delta > 0$; a repeated real one iff $\Delta = 0$; a conjugate pair $\alpha \pm i\beta$ iff $\Delta < 0$. *What complex means, concretely:* a real eigenvalue would give a line that the matrix maps into itself; a complex pair says no such line exists — every direction gets turned. The matrix then acts as a rotation combined with a scaling, and the scaling factor per application is the modulus $|\lambda| = \sqrt{\alpha^2 + \beta^2} = \sqrt{D}$.

Solution.

(a) *First matrix:* $T = 0$, $D = 0 - (-2)(2) = 4$, so $\lambda^2 + 4 = 0$ and $\boxed{\lambda = \pm 2i}$: **complex**. Here $|\lambda| = 2$, and indeed the matrix sends $(1, 0) \mapsto (0, 2)$ and $(0, 1) \mapsto (-2, 0)$: a 90° rotation followed by a scaling by 2. Every direction is turned by a quarter-turn, so no real eigenvector can exist — and that is precisely what the complex answer reports.

Second matrix: $T = 4$, $D = 4 + 1 = 5$, $\Delta = 16 - 20 = -4 < 0$:

$$\lambda = \frac{4 \pm \sqrt{-4}}{2} = 2 \pm i,$$

complex again, with modulus $|\lambda| = \sqrt{4 + 1} = \sqrt{5} = \sqrt{D}$ ✓. This is a rotation by $\arctan(1/2) \approx 26.6^\circ$ together with a scaling by $\sqrt{5} \approx 2.24$ at each application. *Reading for dynamics:* $|\lambda| > 1$, so repeated application spirals *outwards*; $|\lambda| < 1$ would spiral inwards, and $|\lambda| = 1$ would circle forever.

(b) $T = 1 + 1 = 2$ and $D = 1 \cdot 1 - a \cdot (-a) = 1 + a^2$, so

$$\Delta = T^2 - 4D = 4 - 4(1 + a^2) = -4a^2.$$

This is ≤ 0 for every real a , and *strictly* negative exactly when $\boxed{a \neq 0}$. At $a = 0$ the matrix is the identity, whose eigenvalue is 1 (twice, real). So: complex for every nonzero a , however small — an arbitrarily small rotational component already destroys every real eigendirection. There is no gradual transition here.

Common mistake. Reading a complex eigenvalue as an error in the computation, or as “the data are complex”. Nothing is wrong: it is the algebra reporting a rotation. Only the *real* part of the picture — what grows and what shrinks — comes from $|\lambda|$; the imaginary part is a frequency.

Exercise 12. (a) Show $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ has $\lambda = 3$ twice, solve $(A - 3I)\mathbf{v} = \mathbf{0}$, count independent eigenvectors: diagonalizable? (b) Same for $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$. (c) What do (a) and (b) say about “a repeated eigenvalue means not diagonalizable”?

What this exercise uses. $\mathbf{A}$ is **diagonalizable** when there is a basis of eigenvectors — equivalently $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$ with $\mathbf{\Lambda}$ diagonal and the eigenvectors as columns of $\mathbf{P}$; $\mathbf{P}$ is invertible exactly when those columns are independent. So the question is never “how many distinct eigenvalues?” but *how many independent eigenvectors can I actually produce?* For a given λ that number is the dimension of the solution space of $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$, and it can be smaller than the number of times λ repeats.

Solution.

- (a) $T = 3 + 3 = 6$, $D = 3 \cdot 3 - 1 \cdot 0 = 9$, so

$$\lambda^2 - 6\lambda + 9 = (\lambda - 3)^2 = 0 :$$

$\lambda = 3$, a **double** root. Now solve:

$$\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies v_2 = 0,$$

with v_1 free. The solutions are the multiples of $(1, 0)$: a **line**, so **one** independent eigenvector for an eigenvalue repeated twice. Two independent eigenvectors are needed for a basis of $\mathbb{R}^2$ and only one exists, so the matrix is **not diagonalizable** (it is called *defective*).

What it does instead: $\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \mapsto \begin{pmatrix} 3v_1 + v_2 \\ 3v_2 \end{pmatrix}$ — it scales by 3 *and* shears, adding a bit of the second coordinate into the first. A shear has no second eigendirection to offer.

- (b) $\mathbf{A} = 3\mathbf{I}$: $T = 6$, $D = 9$, so again $(\lambda - 3)^2 = 0$ and $\lambda = 3$ twice — *the very same characteristic polynomial as in (a)*. But now

$$\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

so *every* vector solves $(\mathbf{A} - 3\mathbf{I})\mathbf{v} = \mathbf{0}$: the solution space is the whole plane, and there are **two** independent eigenvectors, e.g. $(1, 0)$ and $(0, 1)$ (or any other basis — every direction is an eigendirection). The matrix is already diagonal, hence trivially diagonalizable.

- (c) The claim is **false**. (a) and (b) have identical eigenvalues, identical characteristic polynomials, the same repetition — and opposite answers. What decides is the number of independent eigenvectors attached to the repeated eigenvalue: one in (a), two in (b).

What *is* true is the one-way statement: n *distinct* eigenvalues guarantee diagonalizability (eigenvectors for distinct eigenvalues are automatically independent). A repetition removes the guarantee; it does not deliver a verdict.

Remark. Compare Exercise 19, where a covariance matrix has a repeated eigenvalue with a full plane of eigenvectors — case (b). This is not luck: symmetric matrices are always diagonalizable (Exercise 15), so the defect of case (a) can never occur for a covariance matrix.

Exercise 13. (a) Eigenvalues -2 and 0.5 : applied many times, which direction wins, and what does the sign add? (b) $A = P\Lambda P^{-1}$ with $\lambda_1 = 1$, $\lambda_2 = \frac{1}{2}$: describe A^{10} . (c) Using Exercise 10(b), compute A^3 for $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and check it directly.

What this exercise uses. If $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$ then the inner factors telescope:

$$\mathbf{A}^2 = \mathbf{P}\mathbf{\Lambda}\underbrace{\mathbf{P}^{-1}\mathbf{P}}_{=\mathbf{I}}\mathbf{\Lambda}\mathbf{P}^{-1} = \mathbf{P}\mathbf{\Lambda}^2\mathbf{P}^{-1}, \quad \text{and generally} \quad \mathbf{A}^k = \mathbf{P}\mathbf{\Lambda}^k\mathbf{P}^{-1},$$

where $\mathbf{\Lambda}^k = \text{diag}(\lambda_1^k, \lambda_2^k)$ — powers of a diagonal matrix are taken entry by entry. So iterating a linear rule is just raising each eigenvalue to the power k : $|\lambda| > 1$ **grows**, $|\lambda| < 1$ **decays**, $|\lambda| = 1$ **persists**, and after many steps only the largest $|\lambda|$ is still visible.

Solution.

- (a) Compare moduli: $|-2| = 2$ against $|0.5| = 0.5$. Writing a generic $\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$,

$$\mathbf{A}^k\mathbf{x} = c_1(-2)^k\mathbf{v}_1 + c_2(0.5)^k\mathbf{v}_2,$$

and the ratio of the two magnitudes is $(2/0.5)^k = 4^k$: after 10 steps the first term is about a million times the second. The result therefore lines up with the eigendirection of $\boxed{-2}$ — the *dominant* one, meaning largest in absolute value, not largest as a number.

What the sign adds: $(-2)^k$ alternates in sign, so the vector flips to the other side of the origin at every step. It converges to the **line** spanned by $\mathbf{v}_1$, not to a fixed vector, and its norm doubles each time. Oscillation and growth are two independent pieces of information: the sign and the modulus.

- (b) $\mathbf{A}^{10} = \mathbf{P} \operatorname{diag}(1^{10}, (\frac{1}{2})^{10}) \mathbf{P}^{-1} = \mathbf{P} \operatorname{diag}(1, \frac{1}{1024}) \mathbf{P}^{-1}$. The first mode is *untouched* ($1^{10} = 1$); the second is divided by 1024. Only the $\lambda = 1$ mode survives, so for any starting vector $\mathbf{A}^{10}\mathbf{x}$ is essentially $c_1\mathbf{v}_1$: the state collapses onto a single direction while keeping a constant size along it. (An eigenvalue exactly 1 is the borderline case: neither growth nor decay — a persistent memory.)
- (c) From Exercise 10(b): $\lambda_1 = 3$ with $\mathbf{v}_1 = (1, 1)$, $\lambda_2 = -1$ with $\mathbf{v}_2 = (1, -1)$. So $\mathbf{A}^3$ has the same eigenvectors with eigenvalues $3^3 = 27$ and $(-1)^3 = -1$. Rather than inverting $\mathbf{P}$, use the action on the two eigenvectors and rebuild:

$$\mathbf{A}^3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 27 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 27 \\ 27 \end{pmatrix}, \quad \mathbf{A}^3 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Now $\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2}[\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix}]$, so the first column of $\mathbf{A}^3$ is $\frac{1}{2}[\begin{pmatrix} 27 \\ 27 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \end{pmatrix}] = \frac{1}{2}\begin{pmatrix} 26 \\ 28 \end{pmatrix} = \begin{pmatrix} 13 \\ 14 \end{pmatrix}$; similarly $\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{2}[\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \end{pmatrix}]$ gives the second column $\frac{1}{2}\begin{pmatrix} 28 \\ 26 \end{pmatrix} = \begin{pmatrix} 14 \\ 13 \end{pmatrix}$. Hence

$$\mathbf{A}^3 = \begin{pmatrix} 13 & 14 \\ 14 & 13 \end{pmatrix}.$$

Direct check:

$$\begin{aligned} \mathbf{A}^2 &= \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1+4 & 2+2 \\ 2+2 & 4+1 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}, \\ \mathbf{A}^3 &= \mathbf{A}^2 \mathbf{A} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 5+8 & 10+4 \\ 4+10 & 8+5 \end{pmatrix} = \begin{pmatrix} 13 & 14 \\ 14 & 13 \end{pmatrix}. \checkmark \end{aligned}$$

Sanity check on the trace: $\operatorname{tr} \mathbf{A}^3 = 26 = 27 + (-1) \checkmark$.

Common mistake. Ranking eigenvalues by value instead of by modulus. Between -2 and 0.5 the larger *number* is 0.5 ; the dominant *mode* is -2 . For repeated application, only $|\lambda|$ decides who wins, and the sign decides whether the approach is monotone or alternating.

Exercise 14. A 2×2 matrix has eigenvalues -3 and 5 . Give its trace and determinant without knowing the matrix.

What this exercise uses. Expanding $(\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2$ and comparing with $\lambda^2 - T\lambda + D$ identifies the coefficients:

$$T = \lambda_1 + \lambda_2, \quad D = \lambda_1\lambda_2.$$

Both are therefore *basis-independent*: they depend on the transformation, not on the coordinates used to write it down.

Solution.

$$T = \operatorname{tr} \mathbf{A} = -3 + 5 = \boxed{2}, \quad D = \det \mathbf{A} = (-3) \times 5 = \boxed{-15}.$$

Two readings. $\det = -15 \neq 0$: the matrix is invertible — consistent with the fact that 0 is not an eigenvalue, since $\lambda = 0$ would mean some nonzero vector is sent to $\mathbf{0}$. And $\det < 0$: the two eigenvalues have opposite signs, so the map stretches by 5 along one axis while reflecting and stretching by 3 along the other, reversing orientation overall. Areas are multiplied by 15.

The matrix itself is *not* determined: for example $\begin{pmatrix} -3 & 0 \\ 0 & 5 \end{pmatrix}$ and $\begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$ both have $T = 2$, $D = -15$. (Check the second: $T = 2$, $D = 1 - 16 = -15$ ✓, eigenvalues 5 and -3 .) Infinitely many matrices share a spectrum; what they share is a *behaviour*, in different axes.

Remark. These two identities are the cheapest error-detector in the session: after any eigenvalue computation, verify that the sum is the trace and the product is the determinant. It costs one line and catches almost every arithmetic slip.

Exercise 15. (a) $\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$: symmetric? eigenvalues? are the eigenvectors orthogonal? (b) same for $\begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$. (c) Give a real matrix with complex eigenvalues and explain, without computing, why it cannot be symmetric.

What this exercise uses. $\mathbf{A}$ is **symmetric** when $\mathbf{A}^T = \mathbf{A}$, i.e. $a_{ij} = a_{ji}$: the matrix is unchanged by a reflection in its diagonal. The **spectral theorem** says that a real symmetric matrix always has (i) *real* eigenvalues and (ii) an *orthonormal* basis of eigenvectors — in particular it is always diagonalizable, and $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ with $\mathbf{Q}$ orthogonal, so the inverse in the diagonalisation is a mere transpose. This is the theorem that makes PCA work, since a covariance matrix is symmetric by construction.

Solution.

(a) Symmetric: the off-diagonal entries are both 1. ✓ $T = 4 + 4 = 8$, $D = 16 - 1 = 15$, so

$$\lambda^2 - 8\lambda + 15 = (\lambda - 5)(\lambda - 3) = 0 : \quad \lambda = 5 \text{ and } \lambda = 3,$$

both real, as promised.

$\lambda = 5$: $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$ gives $v_1 = v_2$, so $\mathbf{v}_1 = (1, 1)$.

$\lambda = 3$: $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ gives $v_1 = -v_2$, so $\mathbf{v}_2 = (1, -1)$.

Orthogonality: $(1, 1) \cdot (1, -1) = 1 - 1 = 0$. ✓ Normalised, the orthonormal eigenbasis is $\left\{ \frac{1}{\sqrt{2}}(1, 1), \frac{1}{\sqrt{2}}(1, -1) \right\}$.

(b) Symmetric likewise. $T = 10$, $D = 25 - 4 = 21$:

$$\lambda^2 - 10\lambda + 21 = (\lambda - 7)(\lambda - 3) = 0 : \quad \lambda = 7 \text{ and } \lambda = 3.$$

$\lambda = 7$: $\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$ gives $\mathbf{v}_1 = (1, 1)$; $\lambda = 3$: $\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$ gives $\mathbf{v}_2 = (1, -1)$; orthogonal again ✓.

Pattern worth memorising: every matrix of the form $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ has eigenvector $(1, 1)$ with eigenvalue $a + b$ and eigenvector $(1, -1)$ with eigenvalue $a - b$. Indeed $\begin{pmatrix} a & b \\ b & a \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a+b \\ a+b \end{pmatrix}$ and $\begin{pmatrix} a & b \\ b & a \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} a-b \\ b-a \end{pmatrix}$. Check: (a) $4 \pm 1 = 5, 3$ ✓; (b) $5 \pm 2 = 7, 3$ ✓. This pattern is used again in Exercises 16(b), 18 and 22.

(c) Take the 90° rotation

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} : \quad T = 0, \quad D = 0 - (-1)(1) = 1, \quad \lambda^2 + 1 = 0, \quad \lambda = \pm i.$$

It cannot be symmetric, and no computation is needed to say so: the spectral theorem guarantees that a real symmetric matrix has *only real* eigenvalues, so a single complex eigenvalue rules symmetry out. (Here one can also just look: $r_{12} = -1 \neq 1 = r_{21}$.) The logical form matters — we are using the contrapositive of the theorem, “not real $\Rightarrow$ not symmetric”.

Common mistake. Believing the converse, that non-symmetric implies complex. Exercise 10(c), $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$, is not symmetric and has eigenvalues 5 and 2, perfectly real — but with *non-orthogonal* eigenvectors. Symmetry guarantees real and orthogonal; its absence guarantees nothing.

Exercise 16. (a) A covariance matrix has eigenvalues 8 and 0.5: along which direction does the data spread, and what fraction of the variance is there? (b) For $C = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}$ (two similar tasks): the two principal components in words, the fraction along the first, and the trace check.

What this exercise uses. For centred data, $C = \frac{1}{n} X^T X$ has entry (j, k) equal to $\frac{1}{n} \sum_i X_{ij} X_{ik}$: the variance of measure j on the diagonal, the covariance of j and k off it. It is symmetric, so by Exercise 15 it has real eigenvalues and orthonormal eigenvectors — these are the **principal components**, and λ_k is exactly the variance of the data along $\mathbf{v}_k$. Total variance is $\sum_j \text{Var}(j) = \text{tr } C = \sum_k \lambda_k$, so the *proportion of variance explained* by component k is $\lambda_k / \sum_j \lambda_j$.

Solution.

- (a) The data spread most along the eigenvector of the **largest** eigenvalue, i.e. of $\lambda_1 = 8$. Its share of the total variance is

$$\frac{8}{8 + 0.5} = \frac{8}{8.5} = \frac{16}{17} \approx 0.941 = 94.1\%.$$

The remaining $0.5/8.5 \approx 5.9\%$ lies in the perpendicular direction. The cloud is a strongly elongated ellipse: the ratio of its half-axes is $\sqrt{8}/\sqrt{0.5} = \sqrt{16} = 4$, so it is four times longer than it is wide. (Variances are squared lengths — take square roots to talk about the shape.)

- (b) $T = 10$, $D = 25 - 9 = 16$:

$$\lambda^2 - 10\lambda + 16 = (\lambda - 8)(\lambda - 2) = 0 : \quad \lambda_1 = 8, \quad \lambda_2 = 2,$$

and by the $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ pattern of Exercise 15(b), $\lambda_1 = 5 + 3 = 8$ along $(1, 1)$ and $\lambda_2 = 5 - 3 = 2$ along $(1, -1)$. Normalised: $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$, $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)$.

In words. $\mathbf{v}_1$ weights the two tasks equally and with the same sign: it is the **overall level** on the two tasks — “some participants are good at both, some are weak at both”, and this is where most of the variation is. $\mathbf{v}_2$ opposes them: it is the **relative advantage**, “better at task 1 than at task 2”. This is the same level/contrast pair as Session 1, Exercise 20 — except that here the data *chose* those axes rather than the experimenter.

Fraction along the first:

$$\frac{8}{8 + 2} = \frac{8}{10} = 80\%.$$

Trace check: $\text{tr } C = 5 + 5 = 10$ and $\lambda_1 + \lambda_2 = 8 + 2 = 10$. ✓

Common mistake. Reporting λ_1 as “the length” of the principal axis. λ_1 is a *variance*; the axis length goes with $\sqrt{\lambda_1}$. Mixing the two makes an ellipse look far more elongated than it is — in (b), 8 against 2 is a factor 4 in variance but only 2 in shape.

Exercise 17. Which could be covariance matrices? Justify with eigenvalues. (a) $\begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$ — and which axis is PC1? (b) $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ — and its rank? (c) $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$; (d) $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.

What this exercise uses. A covariance matrix must be symmetric *and* **positive semi-definite**: all eigenvalues ≥ 0 . The reason is not a convention. The variance of the data projected on a unit direction $\mathbf{u}$ is $\mathbf{u}^T C \mathbf{u}$, a genuine variance and hence never negative; taking $\mathbf{u} = \mathbf{v}_k$ an eigenvector gives $\mathbf{v}_k^T C \mathbf{v}_k = \lambda_k$. **So a negative eigenvalue would be a negative variance** — impossible. Zero *is* allowed: it means the data have no spread at all in that direction, i.e. they lie in a lower-dimensional flat.

Solution.

- (a) Diagonal, so the eigenvalues are 4 and 9, both ≥ 0 : **yes**, a valid covariance matrix. The off-diagonal zeros say the two measures are uncorrelated. PC1 is the direction of the largest eigenvalue 9, i.e. the **y-axis**, $(0, 1)$ — the second measure, whose variance is larger. (The naming is by variance, not by column order.)
- (b) $T = 1 + 4 = 5$, $D = 1 \cdot 4 - 2 \cdot 2 = 0$, so $\lambda^2 - 5\lambda = \lambda(\lambda - 5) = 0$: eigenvalues 5 and 0. Both ≥ 0 : **yes**, valid — but degenerate. Since $\det = 0$ and the matrix is nonzero, $\text{rank} = 1$.
What that means for the data: zero variance along the eigenvector of $\lambda = 0$, so *every* participant lies exactly on one line through the origin. The two measures are perfectly correlated ($r = 1$); one is an exact multiple of the other. Indeed $C = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, an outer product (Exercise 6).
- (c) From Exercise 10(b) the eigenvalues are 3 and $\boxed{-1}$. **No**: a negative eigenvalue means a direction along which the “variance” would be -1 . Concretely, the projection on $\frac{1}{\sqrt{2}}(1, -1)$ would have variance -1 — impossible for real data. The matrix is symmetric, so it passes the first test; symmetry alone is not enough.
- (d) $T = 4$, $D = 4 - 1 = 3$: $\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1) = 0$, eigenvalues 3 and 1, both positive: **yes**, and non-degenerate. By the $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ pattern with $b = -1$: $\lambda = 1$ along $(1, 1)$, $\lambda = 3$ along $(1, -1)$. Here the *contrast* direction carries most of the variance — the negative covariance means the two measures trade off against each other.

Remark. Note (c) versus (d): the same symmetric shape, one sign changed off the diagonal, and one is a possible covariance matrix while the other is not. The admissible range is $|b| \leq a$ for $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$, since the eigenvalues are $a \pm b$ — which is the matrix version of $|\text{correlation}| \leq 1$.

Exercise 18. Four centred participants, two measures: $(1, 1)$, $(-1, -1)$, $(2, 2)$, $(-2, -2)$. (a) Compute $C = \frac{1}{4} \sum_i \mathbf{x}_i \mathbf{x}_i^\top$. (b) Eigenvalues and PC1. (c) What does the second eigenvalue tell you?

What this exercise uses. $C = \frac{1}{n} \sum_i \mathbf{x}_i \mathbf{x}_i^\top$ is the same object as $\frac{1}{n} X^\top X$, written as a sum of outer products: each participant contributes one rank-1 matrix $\mathbf{x}_i \mathbf{x}_i^\top$, and the covariance is their average. The formula is only a covariance matrix if the data are *already centred* (mean $\mathbf{0}$) — check here: $(1, 1) + (-1, -1) + (2, 2) + (-2, -2) = (0, 0)$ ✓.

Solution.

- (a) Each term is $\mathbf{x} \mathbf{x}^\top = \begin{pmatrix} x_1^2 & x_1 x_2 \\ x_1 x_2 & x_2^2 \end{pmatrix}$:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} (1 \ 1) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ -1 \end{pmatrix} (-1 \ -1) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

$$\begin{pmatrix} 2 \\ 2 \end{pmatrix} (2 \ 2) = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}, \quad \begin{pmatrix} -2 \\ -2 \end{pmatrix} (-2 \ -2) = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}.$$

(Note that $\mathbf{x}$ and $-\mathbf{x}$ contribute the *same* matrix — the signs cancel in every product.) Summing and dividing by 4:

$$\sum_i \mathbf{x}_i \mathbf{x}_i^\top = \begin{pmatrix} 10 & 10 \\ 10 & 10 \end{pmatrix}, \quad C = \frac{1}{4} \begin{pmatrix} 10 & 10 \\ 10 & 10 \end{pmatrix} = \begin{pmatrix} 2.5 & 2.5 \\ 2.5 & 2.5 \end{pmatrix}.$$

Read off the entries: both measures have variance 2.5, and their covariance is also 2.5 — as large as it can possibly be, which already announces the answer to (c).

(b) $T = 5$, $D = 2.5^2 - 2.5^2 = 0$, so $\lambda^2 - 5\lambda = 0$ and

$$\lambda_1 = 5, \quad \lambda_2 = 0.$$

By the $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ pattern with $a = b = 2.5$: $\lambda_1 = a + b = 5$ along $(1, 1)$ and $\lambda_2 = a - b = 0$ along $(1, -1)$. So

$$\text{PC1} = \frac{1}{\sqrt{2}}(1, 1),$$

with $5/5 = 100\%$ of the variance.

(c) $\lambda_2 = 0$ means **exactly zero spread** in the direction $(1, -1)$: the two measures are perfectly redundant. Geometrically the four points $(1, 1), (-1, -1), (2, 2), (-2, -2)$ all sit on the line $y = x$, so the data are genuinely *one*-dimensional even though they were recorded with two numbers. One measure could be deleted with no loss at all.

This is the same phenomenon as Exercise 5, seen from the covariance side rather than the rank side: $\text{rank } C = 1$, and a zero eigenvalue of C is a direction in which nothing happens.

Exercise 19. Now the four centred points are $(2, 0), (0, 2), (-2, 0), (0, -2)$. Compute C and say what is special about its principal components.

Solution. The four outer products are

$$\begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}, \quad \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix},$$

so

$$\sum_i \mathbf{x}_i \mathbf{x}_i^T = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}, \quad C = \frac{1}{4} \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2\mathbf{I}.$$

(Mean check: the four points sum to $(0, 0)$, so they are centred. ✓)

What is special. $T = 4$, $D = 4$, $\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2$: the eigenvalue 2 is **double**. And since $C - 2\mathbf{I} = \mathbf{0}$, *every* nonzero vector is an eigenvector — there is a whole plane of them, not two lines. Consequences, all the same statement:

- the cloud is **isotropic**: the variance is 2 in every direction, so it is a disc, not an ellipse, and no direction is preferred;
- PCA has *nothing to choose*. Any orthonormal pair of directions is a legitimate pair of principal components;
- consequently a numerical routine will return *some* pair of axes — and those axes are an artefact of the algorithm and of rounding, not a property of the data. Two runs on slightly perturbed data can return wildly different axes, both correct.

Remark. Compare Exercise 12(b): a repeated eigenvalue with a full plane of eigenvectors, exactly the non-defective case. Symmetric matrices can only repeat eigenvalues in this benign way. The practical warning is nonetheless real: whenever two successive eigenvalues of a covariance matrix are close, the corresponding components are individually unstable, and only the plane they span is meaningful. Do not interpret PC2 and PC3 separately when $\lambda_2 \approx \lambda_3$.

Exercise 20. X is a centred 200×6 table, $X = U\Sigma V^T$. (a) Dimensions of U, Σ, V and how many singular values? (b) Which factor holds the directions in measure space, which has one row per participant? (c) Why not take eigenvalues of X directly?

What this exercise uses. The **SVD**: every real $n \times p$ matrix factors as $X = U\Sigma V^T$ with U orthogonal $n \times n$, V orthogonal $p \times p$, and Σ an $n \times p$ “diagonal” matrix carrying $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ on its main diagonal and zeros elsewhere. Read right to left, the recipe is: *rotate in measure space (V^T)*, *stretch each new axis by its σ_k (Σ)*, *rotate in participant space (U)*. Unlike an eigendecomposition it exists for *every* matrix, square or not.

Solution.

- (a) With $n = 200$ and $p = 6$:

$$U : 200 \times 200, \quad \Sigma : 200 \times 6, \quad V : 6 \times 6.$$

Dimension check: $(200 \times 200)(200 \times 6)(6 \times 6) = 200 \times 6$ ✓. The number of singular values is $\min(n, p) = \boxed{6}$ — the diagonal of a 200×6 matrix has only 6 entries; the other 194 rows of Σ are entirely zero.

(In practice one stores the *thin* SVD: the first 6 columns of U and a 6×6 Σ . The remaining 194 columns of U multiply nothing but zeros.)

- (b) V holds the directions in measure space: its columns $\mathbf{v}_k$ have one coordinate per measure (6 of them), so they are vectors of $\mathbb{R}^6$ — these are the **principal components**.

U has one row per participant (200 rows). The component *scores* — where each participant sits along each component — are the columns of $U\Sigma$, i.e. $\sigma_k \mathbf{u}_k$; equivalently $XV = U\Sigma$.

The mnemonic is dimensional and cannot be got wrong: a vector of measure space must have 6 coordinates, and only the columns of V do.

- (c) Because $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ requires input and output to live in the *same* space, which forces $\mathbf{A}$ to be square. For X of size 200×6 , $X\mathbf{v}$ takes a vector of $\mathbb{R}^6$ and returns one of $\mathbb{R}^{200}$; the equation $X\mathbf{v} = \lambda\mathbf{v}$ compares a 6-vector with a 200-vector and is not even well-formed. Eigenvalues of a data table are simply **undefined**.

The SVD is the repair: it gives two *different* orthonormal bases, one on each side, related by $X\mathbf{v}_k = \sigma_k \mathbf{u}_k$. That is the honest analogue of an eigen-equation for a rectangular matrix. And symmetric square matrices are recovered as the special case $U = V$.

Exercise 21. From $X = U\Sigma V^T$ with $U^T U = I$, $V^T V = I$: (a) show $XX^T = U(\Sigma\Sigma^T)U^T$ and say of what U is the eigenvector matrix; (b) why do $X^T X$ and XX^T have the same nonzero eigenvalues although $\Sigma^T \Sigma$ and $\Sigma\Sigma^T$ have different sizes? (c) X is 200×6 : sizes of $X^T X$ and XX^T , and how many eigenvalues of XX^T must vanish.

What this exercise uses. Two rules do all the work: $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ (the order reverses) and $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$ for an orthogonal $\mathbf{Q}$. In the lecture the same computation on the other side gave $X^T X = V(\Sigma^T \Sigma)V^T$, showing that the columns of V are the eigenvectors of $X^T X$ with eigenvalues σ_k^2 — which is exactly why the principal components of $C = \frac{1}{n} X^T X$ are the right singular vectors.

Solution.

- (a) Transpose first, using the reversal rule twice:

$$X^T = (U\Sigma V^T)^T = (V^T)^T \Sigma^T U^T = V \Sigma^T U^T.$$

Now multiply:

$$XX^T = (U\Sigma V^T)(V \Sigma^T U^T) = U \Sigma \underbrace{(V^T V)}_{=I} \Sigma^T U^T = U(\Sigma\Sigma^T)U^T.$$

Since $\Sigma\Sigma^\top$ is diagonal and U is orthogonal ($U^{-1} = U^\top$), this *is* an eigendecomposition: the columns of U are the eigenvectors of $\boxed{XX^\top}$, with eigenvalues the diagonal entries σ_k^2 (and zeros).

What that matrix is. $X^\top X$ is $p \times p$ and compares **measures** with each other; $XX^\top$ is $n \times n$ and compares **participants** with each other — its entry (i, i') is $\mathbf{x}_i \cdot \mathbf{x}_{i'}$, the similarity of two participants' profiles. So V describes measure space and U describes participant space, exactly as Exercise 20(b) said.

- (b) $\Sigma^\top\Sigma$ is $p \times p$ and $\Sigma\Sigma^\top$ is $n \times n$; both are diagonal, and both carry the *same* numbers $\sigma_1^2, \dots, \sigma_{\min(n,p)}^2$ on the diagonal. The larger one simply pads with extra zeros. Since eigenvalues of a diagonal matrix are its diagonal entries, $X^\top X$ and $XX^\top$ have identical *nonzero* spectra and differ only in how many zeros they carry.

The intuition: σ_k^2 measures how much of the data lies along mode k , and that quantity does not care whether you look at it from the measure side or the participant side. It is one number describing one mode, viewed twice.

(c)

$$X^\top X : (6 \times 200)(200 \times 6) = \boxed{6 \times 6}, \quad XX^\top : (200 \times 6)(6 \times 200) = \boxed{200 \times 200}.$$

$XX^\top$ has 200 eigenvalues, of which at most 6 are nonzero (there are only 6 singular values), so **at least 194 are zero**. Equivalently $\text{rank}(XX^\top) = \text{rank } X \leq 6$.

This is why nobody ever forms $XX^\top$ when $n \gg p$: it is a 200×200 matrix that is almost entirely redundant, when the 6×6 matrix $X^\top X$ holds all the information. (When $p \gg n$ — more voxels than scans, more genes than patients — the calculation is reversed and $XX^\top$ is the small one. This is the “kernel trick” of a lot of neuroimaging software.)

Common mistake. Writing $(U\Sigma V^\top)^\top = U^\top \Sigma^\top V$. The transpose reverses the order of *every* factor: $V\Sigma^\top U^\top$. Keep the dimension check as a guard — here $(6 \times 6)(6 \times 200)(200 \times 200)$ is legal and 6×200 as it should be.

Exercise 22. $X = \begin{pmatrix} 3 & 1 \\ -3 & -1 \\ -1 & -3 \end{pmatrix}$ (four centred participants, two measures). (a) $X^\top X$, its eigenvalues, the singular values. (b) $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{u}_k = X\mathbf{v}_k/\sigma_k$, with the orthonormality checks. (c) Fraction of variance on the first component. (d) The best rank-1 approximation X_1 , the error and its rank, and the check $X = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^\top$.

What this exercise uses. The recipe for computing an SVD by hand, all four steps: **(1)** form $X^\top X$ (small, symmetric, positive semi-definite); **(2)** its eigenvalues are σ_k^2 , so $\sigma_k = \sqrt{\lambda_k}$ — take the positive root, singular values are ≥ 0 by convention; **(3)** its orthonormal eigenvectors are the $\mathbf{v}_k$; **(4)** recover $\mathbf{u}_k = X\mathbf{v}_k/\sigma_k$, which is legitimate for every $\sigma_k \neq 0$ and automatically produces unit, mutually orthogonal vectors: $\mathbf{u}_j \cdot \mathbf{u}_k = \frac{1}{\sigma_j \sigma_k} \mathbf{v}_j^\top X^\top X \mathbf{v}_k = \frac{\sigma_k^2}{\sigma_j \sigma_k} \mathbf{v}_j \cdot \mathbf{v}_k$, which is 1 if $j = k$ and 0 otherwise.

Solution.

- (a) The columns are $\mathbf{c}_1 = (3, 1, -3, -1)$ and $\mathbf{c}_2 = (1, 3, -1, -3)$, and each entry of $X^\top X$ is a dot product of two columns:

$$\mathbf{c}_1 \cdot \mathbf{c}_1 = 9 + 1 + 9 + 1 = 20, \quad \mathbf{c}_1 \cdot \mathbf{c}_2 = 3 + 3 + 3 + 3 = 12, \quad \mathbf{c}_2 \cdot \mathbf{c}_2 = 1 + 9 + 1 + 9 = 20,$$

$$X^\top X = \begin{pmatrix} 20 & 12 \\ 12 & 20 \end{pmatrix}.$$

Symmetric ✓ (it always is: $(X^T X)^T = X^T X$). By the $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ pattern,

$$\lambda_1 = 20 + 12 = 32, \quad \lambda_2 = 20 - 12 = 8.$$

(Long way, same answer: $T = 40$, $D = 400 - 144 = 256$, $\lambda = \frac{40 \pm \sqrt{1600 - 1024}}{2} = \frac{40 \pm 24}{2}$ ✓.) Hence

$$\sigma_1 = \sqrt{32} = 4\sqrt{2} \approx 5.657, \quad \sigma_2 = \sqrt{8} = 2\sqrt{2} \approx 2.828.$$

(b) From the same pattern,

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1), \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)$$

(normalised, and orthogonal to each other). Now

$$X\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 3+1 \\ 1+3 \\ -3-1 \\ -1-3 \end{pmatrix} = \frac{4}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \quad \mathbf{u}_1 = \frac{X\mathbf{v}_1}{\sigma_1} = \frac{4/\sqrt{2}}{4\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} = \frac{1}{2}(1, 1, -1, -1),$$

$$X\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 3-1 \\ 1-3 \\ -3+1 \\ -1+3 \end{pmatrix} = \frac{2}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 = \frac{X\mathbf{v}_2}{\sigma_2} = \frac{2/\sqrt{2}}{2\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{2}(1, -1, -1, 1).$$

Checks: $\|\mathbf{u}_1\|^2 = \frac{1}{4}(1+1+1+1) = 1$ ✓, $\|\mathbf{u}_2\|^2 = 1$ ✓, and

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = \frac{1}{4}(1 \cdot 1 + 1 \cdot (-1) + (-1)(-1) + (-1)(1)) = \frac{1}{4}(1 - 1 + 1 - 1) = 0. \checkmark$$

(c) Variance goes with the *squares*:

$$\frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \frac{32}{32 + 8} = \frac{32}{40} = 80\%,$$

leaving 20% on the second. *Reading:* $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$ is the “high on both measures” direction and $\mathbf{v}_2$ the “better at measure 1 than at measure 2” contrast; the data split 80/20 between them.

(d)

$$X_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T = 4\sqrt{2} \cdot \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} (1 \quad 1) = \underbrace{\frac{4\sqrt{2}}{2\sqrt{2}}}_{=2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} (1 \quad 1) = \begin{pmatrix} 2 & 2 \\ 2 & 2 \\ -2 & -2 \\ -2 & -2 \end{pmatrix}.$$

The error:

$$X - X_1 = \begin{pmatrix} 3-2 & 1-2 \\ 1-2 & 3-2 \\ -3+2 & -1+2 \\ -1+2 & -3+2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ -1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} (1 \quad -1),$$

an outer product, hence of **rank** 1 (Exercise 6). And it is not any rank-1 matrix but exactly the second term:

$$\sigma_2 \mathbf{u}_2 \mathbf{v}_2^T = 2\sqrt{2} \cdot \frac{1}{2}(1, -1, -1, 1)^T \cdot \frac{1}{\sqrt{2}}(1 \quad -1) = 1 \cdot \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} (1 \quad -1) = \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ -1 & 1 \\ 1 & -1 \end{pmatrix} = X - X_1. \checkmark$$

Therefore

$$X = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^\top,$$

the full SVD written as a sum of rank-1 layers, largest first. Adding all the layers is exact; stopping early is the best approximation of that rank (Eckart–Young).

Common mistake. Reporting $\sigma_1 = 32$. The eigenvalues of $X^\top X$ are the *squares* of the singular values — $\sigma_1 = \sqrt{32}$. The same confusion turns an 80% variance share into a 71% one; Exercise 23(c) is that mistake in its natural habitat.

Exercise 23. $\sigma = (6, 3, 2, 1)$. (a) Variance fraction of the first component? of the first two? (b) The best rank-2 approximation X_2 and the rank of the error. (c) A colleague computes “75%” using σ instead of σ^2 : recompute and explain.

What this exercise uses. The total variance of a centred table is $\sum_k \sigma_k^2$ (this is the sum of the eigenvalues of $X^\top X$, i.e. $\text{tr}(X^\top X)$, i.e. the sum of all squared entries of X). Component k carries σ_k^2 of it, so the share of the first r components is $\sum_{k \leq r} \sigma_k^2 / \sum_k \sigma_k^2$. **Eckart–Young:** truncating the SVD after r terms, $X_r = \sum_{k \leq r} \sigma_k \mathbf{u}_k \mathbf{v}_k^\top$, is the best rank- r approximation of X there is.

Solution.

(a) Square first:

$$\begin{aligned} \sigma^2 &= (36, 9, 4, 1), & \text{total} &= 36 + 9 + 4 + 1 = 50. \\ \text{first: } \frac{36}{50} &= 72\%, & \text{first two: } \frac{36 + 9}{50} &= \frac{45}{50} = 90\%. \end{aligned}$$

(b)

$$X_2 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^\top,$$

and the error is what was dropped,

$$X - X_2 = \sigma_3 \mathbf{u}_3 \mathbf{v}_3^\top + \sigma_4 \mathbf{u}_4 \mathbf{v}_4^\top,$$

a sum of two rank-1 terms built on *independent* (indeed orthogonal) vectors, hence of **rank 2**. Its share of the variance is $(4 + 1)/50 = 10\%$, complementing (a) ✓.

(c) Their computation was

$$\frac{6 + 3}{6 + 3 + 2 + 1} = \frac{9}{12} = 75\%.$$

It is wrong because **variance is a squared quantity**: σ_k is in the units of the data (a length), σ_k^2 is in squared units (a variance), and shares must be taken of the thing that adds up to the total variance. The correct figure is 90%.

Direction of the error, worth noting: the wrong recipe gives 75% against a true 90%, i.e. it *understates* how concentrated the data are. Squaring exaggerates the gaps between singular values, so the big components gain and the small ones lose; averaging the unsquared values flattens the noise directions.

Common mistake. The one in (c) — and it is common because software often prints the singular values, not their squares. Whenever you see a scree plot, check whether the vertical axis is σ_k or σ_k^2 ; only the second is a variance, and only the second is what percentages are computed from.

Exercise 24. 20 items, 500 people; the singular values of the centred table drop sharply after the second. (a) Roughly what rank, and roughly what *effective* rank? Same question? (b) How many

numbers does the rank-2 truncation need, against 10 000? (c) In one sentence, what is the truncation *not* entitled to claim?

What this exercise uses. **Rank** is exact and algebraic: the number of strictly nonzero singular values. **Effective rank** is a judgement: the number of σ_k big enough to be worth keeping — where the scree plot has its elbow. Real data have measurement noise in every direction, so every σ_k is strictly positive and the exact rank is maximal; the interesting quantity is nevertheless the effective one.

Solution.

- (a) **Exact rank:** 20, almost surely. $\text{rank} \leq \min(500, 20) = 20$, and equality holds unless some column is an *exact* linear combination of the others — which noisy measurements essentially never are (unlike Exercise 5, where the redundancy was built in by construction).

Effective rank: 2, since the singular values collapse after the second: two directions carry the signal, the remaining 18 carry noise.

These are not the same question. The first has a yes/no answer produced by an algorithm; the second is a modelling decision — it depends on the noise level, on the sample size, on what the components will be used for, and two careful analysts may reasonably disagree. Reporting a rank of 20 is true and useless; reporting 2 is useful and requires justification.

- (b) The truncation stores, for each of the two kept components, one vector $\mathbf{u}_k$ of length 500, one vector $\mathbf{v}_k$ of length 20, and the number σ_k :

$$2 \times (500 + 20) + 2 = 2 \times 520 + 2 = 1042 \quad \text{numbers,}$$

against $500 \times 20 = 10\,000$ — about a tenth, and the discarded tenth is mostly noise. (In practice one absorbs σ_k into $\mathbf{u}_k$ and stores 1040.)

- (c) That the two components correspond to two real, interpretable psychological constructs. The SVD returns the directions of *maximal variance* under an orthogonality constraint, which is a far more modest claim: the components are a good coordinate system, not necessarily a set of causes, and any rotation of the retained plane fits the data exactly as well.

Remark. The last point is the honest summary of the whole session. Rank, eigenvalues and the SVD tell you how many directions the data have and which ones carry the variance. What those directions mean is not in the algebra — it comes from the design, the items, and the theory, and it is where the responsibility of the researcher begins.
